



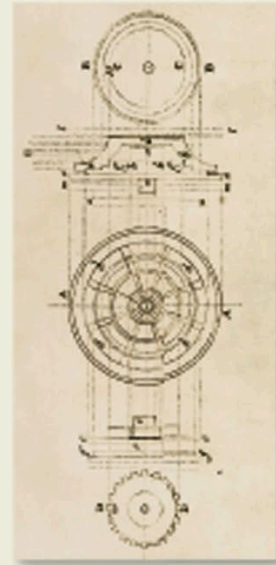
Log float with gauge, c 1861

Measuring speed

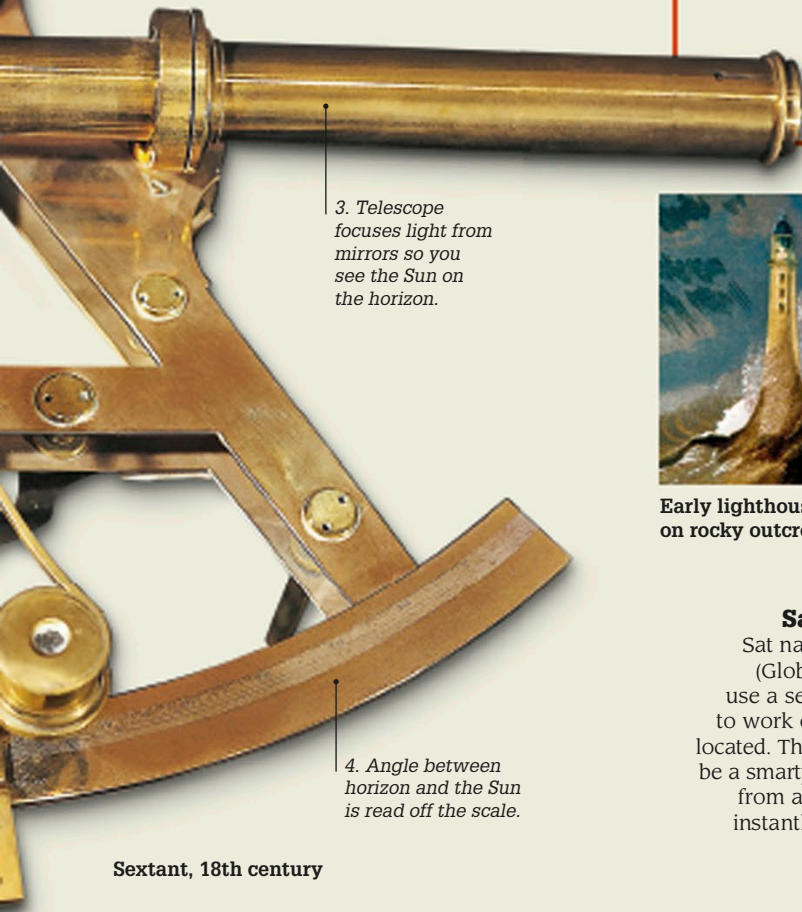
Speed at sea is measured in knots because sailors used to throw a rope (also called a log) tied with knots behind their ship. They allowed it to roll out for a specific amount of time, then counted the number of knots to calculate how fast the ship was moving. Later, a log float with a mechanical gauge was thrown from the ship.

The marine chronometer

In the 1700s, ships were frequently lost at sea because captains had no way of keeping track of longitude (how far the ship has traveled east or west). To do this required an accurate clock that would always show the right time back at home port. Sailors worked out their longitude by comparing that time with "local" time. This was difficult to achieve on board a rolling ship. After working on the problem for many years, John Harrison built the first accurate marine chronometer (timepiece) in 1759.



The original designs for Harrison's chronometer. It was about the size of a large pocket watch.



3. Telescope focuses light from mirrors so you see the Sun on the horizon.

4. Angle between horizon and the Sun is read off the scale.

Sextant, 18th century



Early lighthouses were often built on rocky outcrops.

Lighthouses

The Romans built the first lighthouses at the entrance to harbors. Modern lighthouses date from the early 1800s. They were designed to prevent shipwrecks by shining a powerful beam of light to warn approaching ships of dangerous hazards such as rocky reefs.

Satellite navigation

Sat nav systems such as GPS (Global Positioning System) use a series of global satellites to work out where a receiver is located. The receiver, which could be a smartphone, picks up signals from at least four satellites to instantly calculate its position and speed (see p.232).



1750

Sextants, navigational instruments used to measure the altitude of the planets and stars, had come into use. Sailors used them to determine latitude at night as well as day.

1759

John Harrison built an accurate marine chronometer to calculate longitude at sea.

1935

Radar, which locates unseen objects by bouncing radio waves off them, was invented. It is used on ships to reveal coastlines and other ships.

1990

GPS, the first sat nav system, came into use. It quickly replaced most other navigation aids.

Radar screen

